# Qiang Ma

Multimodal learning / Vision-language models / Medical AI
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## Education

**The University of Hong Kong** 09/2026 - 06/2030 (expected)
*PhD Student in Computer Science; Advisor: Lequan Yu*

**University of Electronic Science and Technology of China** 09/2023 - 06/2026
*MS in Computer Science; Advisor: Xiaoshuang Shi*

**Harbin Engineering University** 09/2018 - 06/2022
*BS in Computer Science*

## Research Experience

**The University of Hong Kong** 09/2025 - 08/2026
*Research Assistant; Advisor: Lequan Yu* Hong Kong

- Proposed PRISM for efficient 3D medical vision-language classification, using spatio-anatomical evidence scoring and foveated token routing to preserve diagnostic regions and compress redundant context.
- Evaluated on bone tumor CT and cardiac MRI with only **15.6%–18.8% of the original visual tokens**, achieving strong classification performance with improved efficiency. [6]

**West China Biomedical Big Data Center, West China Hospital** 08/2024 - 08/2025
*Research Intern; Advisor: Huan Song* Chengdu, China

- Proposed AURA for cross-center prediction with missing modalities at inference, using epistemic uncertainty to guide routing and calibration for more reliable predictions.
- Validated on **five datasets across the US, China, and Korea**, demonstrating superior performance in cross-center transfer and missing-modality settings. [4]

## Publications and Manuscripts

* Co-first authors

[1] **NeuroCogMap Reveals Cognitive Organization of Large Language Models**
Zhongxiang Sun*, Haolang Lu*, **Qiang Ma**, Qi Li, Qipeng Wang, Liang Pang, Chenyu Liu, Qiankun Li, Hao Sun, Kun Wang, Yi Zeng, Jun Xu, Guoqi Li, Ji-Rong Wen.
*Nature*, Under Review, 2026. arXiv:2607.00397.

[2] **Diagnostic Evidence-Based Prototype Learning via Dirichlet Process for Multimodal Bone Tumor Subtyping**
Feng Wu, Ruiling She, Ruiyang Zhang, Liang Peng, **Qiang Ma**, Yinghua Zhao, Lequan Yu.
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026.

[3] **Interpretable Multi-View Fusion Network for Alzheimer's Disease Diagnosis with Large-scale Pre-trained Vision-Language Model**
**Qiang Ma**, Jinghao Xu, Tengfei Li, Xin Yuan, Xiaofeng Zhu, Xiaoshuang Shi.
IEEE International Conference on Bioinformatics and Biomedicine (BIBM), 2025.

[4] **Robust Multimodal Fusion via Epistemic Uncertainty Quantification for Cross-Center Medical Prediction**
**Qiang Ma**, Ziyuan Qin, He Lyu, Junren Wang, Huan Song, Xiaoshuang Shi.
*Information Fusion*, Under Review.

[5] **Shapley-Inspired Causal Credit Assignment for Reinforcement Learning in Efficient Reasoning**
Ziyuan Qin*, **Qiang Ma***, Haoyu Wang, Huahui Yi, Lequan Yu.
Under Review.

[6] **PRISM: Optimizing 3D Medical VLLMs for Classification via Spatio-Anatomical Evidence Purification**
**Qiang Ma**, Feng Wu, Huahui Yi, Ruiyang Zhang, Liang Peng, Shengkun Peng, Xiaoshuang Shi, Lequan Yu.
Under Review.

## Teaching

**Teaching Assistant:** COMP2501 Introduction to Data Science, The University of Hong Kong.

## Academic Service

**Conference and challenge reviewer:** WACV 2027; MICCAI FLARE Challenge 2026; ISBI 2026.

**Journal reviewer:** Pattern Recognition